

Non-Functional Requirements (NFR)

Organization Management Platform

Version 1.0

1. Purpose

This document defines the quality, reliability, security, scalability, maintainability, and operational requirements of the Organization Management Platform.

The goal is to ensure the platform remains stable, secure, maintainable, and extensible as the organization grows.

2. System Overview

The platform includes:

- User Management
- Departments
- Teams
- Roles & Permissions
- Projects
- Tasks
- Subtasks
- Checklists
- Workflows
- Approvals
- Attendance
- Time Tracking
- Documents
- Notifications
- Analytics
- Dashboards
- Automation
- Search
- Audit Logs
- Integrations

Architecture:

- Modular Monolith
 - Domain Driven Design (DDD)
 - Event Driven Core
 - PostgreSQL
 - Redis
-

3. Availability Requirements

System Availability

Target Availability:

99.5%

Allowed Downtime:

Less than 4 hours per month

Recovery

System should recover automatically whenever possible.

Unexpected crashes must not cause data loss.

4. Reliability Requirements

Data Integrity

No operation should leave data in a partially completed state.

Examples:

- Project creation
- Task assignment
- Attendance marking
- Document approval

All critical operations must use database transactions.

Consistency

All users should see consistent data.

No duplicate tasks.

No duplicate approvals.

No orphan records.

5. Performance Requirements

Page Load

Target:

Under 2 seconds

Maximum:

Under 5 seconds

Search

Global search results:

Under 1 second

Dashboard Loading

Dashboard:

Under 3 seconds

Analytics Loading

Standard analytics:

Under 5 seconds

Complex reports:

Under 15 seconds

6. Scalability Requirements

Initial Target:

50-500 users

Future Target:

1000+ users without major redesign

System must support:

- Multiple departments
 - Multiple teams
 - Cross-department projects
 - Multiple managers per user
 - Multiple teams per user
-

7. Security Requirements

Authentication

Required:

- Secure login
- Password hashing
- Session management
- JWT authentication
- Refresh tokens

Optional Future:

- MFA
-

Authorization

Access must be permission-based.

No automatic hierarchy-based access.

Every request must pass permission validation.

Confidential Data

Sensitive information:

- Attendance
- Documents
- Analytics
- Approvals
- User information

Must only be visible to authorized users.

8. Permission Model

Model:

RBAC + Delegated Permissions

Roles:

- Super Admin
- Admin
- Custom Roles

Permissions:

Granted explicitly.

No access should be inferred from job title.

9. Audit Requirements

Every important action must be logged.

Examples:

- Login
- Logout
- Project Creation
- Project Deletion
- Task Assignment
- Attendance Changes
- Document Approval
- Permission Changes

Audit Log Fields:

- User
- Action
- Timestamp
- Old Value
- New Value

Audit records should never be deleted.

10. Search Requirements

Search Scope:

- Projects
 - Tasks
 - Documents
 - Comments
 - Users
 - Teams
 - Departments
 - Workflows
 - Approvals
-

Search Security

Search results must respect permissions.

Users must never see unauthorized data.

Search Protection

Implement:

- Query length limits
 - Rate limiting
 - Input validation
 - SQL injection protection
 - Audit logging
-

11. Backup Requirements

Database Backups:

Daily

Retention:

30 Days

Document Backups:

Daily

Retention:

30 Days

Restore testing:

Monthly

12. Error Handling

System must never expose:

- SQL errors
- Stack traces
- Internal server details

Users should receive friendly error messages.

Errors must be logged internally.

13. Notification Requirements

Supported:

- In-app notifications
- Email notifications

Future:

- Mobile push notifications

Notifications must not block business operations.

14. Document Management Requirements

Documents must support:

- Upload
- Download
- Versioning
- Permissions
- Audit tracking

No permanent deletion without approval.

15. Workflow Requirements

Workflow Engine must support:

- Custom statuses
- Custom transitions
- Approval steps
- Conditional routing

Workflow changes must be versioned.

Existing workflows should not break when new versions are created.

16. Analytics Requirements

Analytics Types:

- Personal Analytics
- Team Analytics
- Department Analytics
- Organization Analytics

Access controlled by permissions.

Analytics calculations must not impact transactional performance.

17. Automation Requirements

Support:

- Trigger
- Condition
- Action

Examples:

Task Completed → Notify Manager

Document Approved → Create Task

Leave Approved → Update Attendance

Automation failures must be logged.

18. Logging Requirements

Log:

- API Requests
- Errors
- Security Events
- Permission Denials
- Automation Failures

Log retention:

90 Days

19. Development Standards

Required:

- Clean Architecture
- Domain Driven Design
- Modular Boundaries
- Code Reviews
- Unit Tests
- Integration Tests

No business logic inside controllers.

No direct database access from UI.

20. Testing Requirements

Coverage Goals:

- Unit Tests: 80%+
- Integration Tests: Critical Flows
- Permission Tests: Mandatory
- Approval Tests: Mandatory

Critical workflows must be tested before release.

21. Deployment Requirements

Environments:

- Development
- Staging
- Production

Production deployments must support rollback.

Database migrations must be reversible whenever possible.

22. Future Expansion Requirements

The architecture must support future additions without major redesign:

- New Modules
- New Dashboards
- New Analytics
- New Workflows
- New Integrations
- Mobile Applications
- Desktop Applications

The system must be designed for evolution rather than perfection.

Final Principle

The platform should prioritize:

1. Data Integrity
2. Security
3. Maintainability
4. Extensibility
5. Performance
6. User Experience

A feature-rich system that loses data or exposes information is considered a failure regardless of functionality.